Foundations of AI:

***Philosophy*:**

Logic, reasoning, mind as a physical system, foundations of learning, language and rationality.

- Where does knowledge come from?
- How does knowledge lead to action?
- How does mental mind arise from physical brain?
- Can formal rules be used to draw valid conclusions?

***Mathematics*:**

Formal representation and proof algorithms, computation, undecidability, intractability, probability.

- What are the formal rules to draw the valid conclusions?
- What can be computed?
- How do we reason with uncertain information?

***Psychology*:**

Adaptation, phenomena of perception and motor control.

- How humans and animals think and act?

***Economics*:**

Formal theory of rational decisions, game theory, operation research.

- How should we make decisions so as to maximize payoff?
- How should we do this when others may not go along?
- How should we do this when the payoff may be far in future?

***Linguistics*:**

Knowledge representation, grammar

- How does language relate to thought?

***Neuroscience*:**

Physical substrate for mental activities

- How do brains process information?

***Control theory*:**

Homeostatic systems, stability, optimal agent design

- How can artifacts operate under their own control?

What is Weak AI?

Weak AI, also known as narrow AI, is artificial intelligence with limited functionality. Weak AI refers to the use of advanced algorithms to accomplish specific problem solving or reasoning tasks that do not encompass the full range of human cognitive abilities. For example, the voice-based personal

assistants such as Siri and Alexa could be considered weak AI programs because they operate within a limited pre-defined set of functions meaning they often have a programmed response. Weak AI is not so enthusiastic about the outcomes of AI; it is simply the view that intelligent behavior can be modeled and used by machines to solve complex problems and tasks. But just because a machine can behave intelligently does not prove that it is actually smart in a way that a human is. The best example of weak AI is Siri and Alexa, or Google Search.

What is Strong AI?

Strong AI is a theoretical form of machine intelligence which supports the view that machines can really develop human consciousness equal to human beings. Strong AI refers to machines or programs with the mind of their own and which can think and accomplish complex tasks on their own without any human interference. Strong AI has complex algorithm that helps systems act in different situations and the strong AI-powered machines can make independent decisions without human interaction. Strong AI-powered machines can carry complex tasks on their own just like human beings do. It simply states that a computing machine with the appropriate functional organization has a mind that perceives, thinks, and intends like a human mind. This is AI we see in sci-fi movies like “Her”, “The Terminator”, “I-Robot”, “WALL-E”and more.